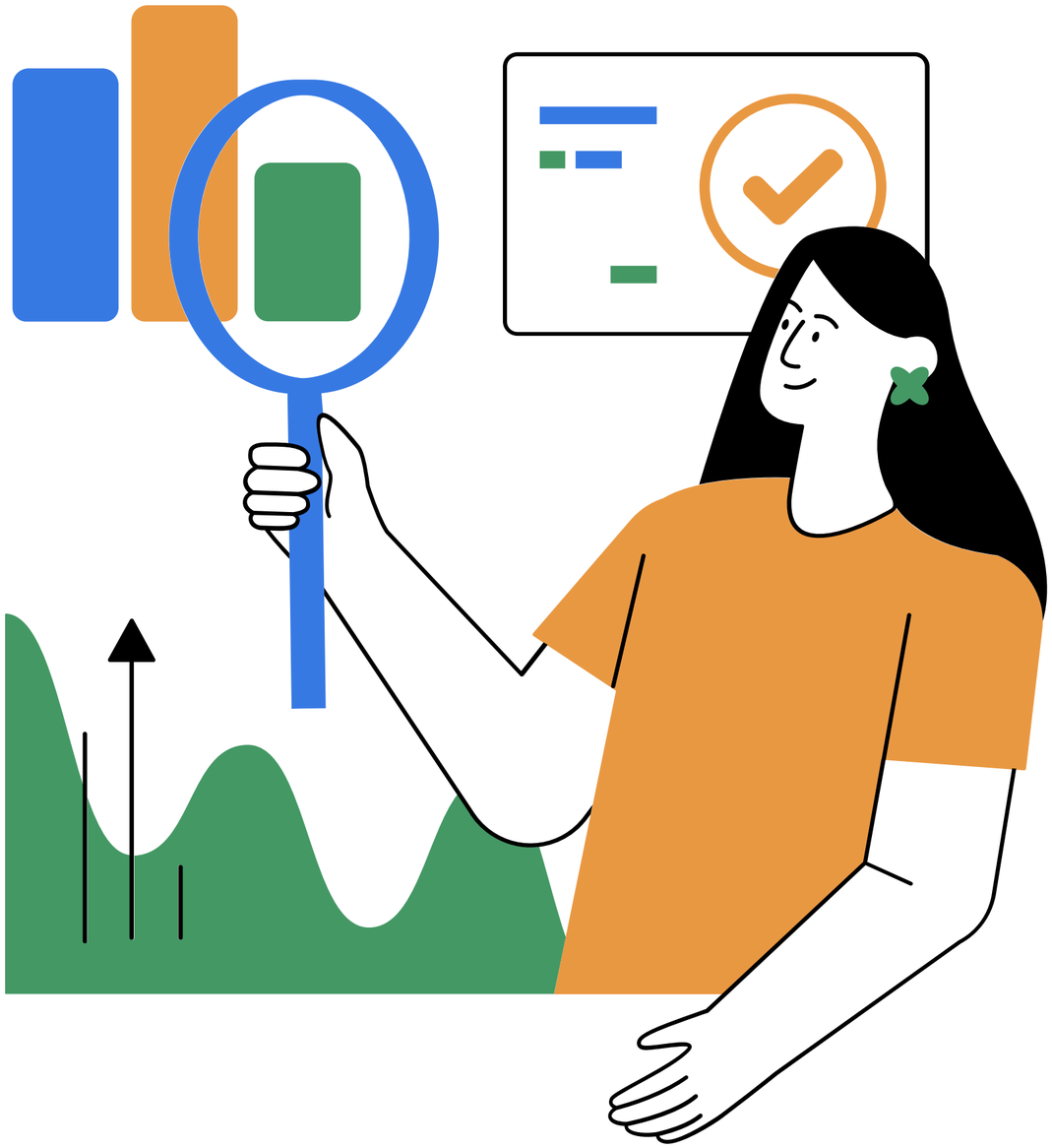
EDA Case Study



Introduction

Analysis Explored

Q1 Selection of High Impact Driving Forces (exploratory)
Q2 Cleaning up Data
Q3 Categorical Data Analysis
Q4 Numerical Data Analysis (Heatmap)
Q5 Historical Data Analysis

Selection of High Impact Driving Forces

TARGET	AMT_INCOME_TOTAL	AMT_CREDIT	AMT_ANNUITY	AMT_GOODS_PRICE	REGION_RATING_CLIENT
	DAYS_BIRTH	DAYS_EMPLOYED	NAME_INCOME_TYPE	NAME_EDUCATION_TYPE	DEF_30_CNT_SOCIAL_CIRCLE
	NAME_FAMILY_STATUS	NAME_HOUSING_TYPE	OCCUPATION_TYPE	EXT_SOURCE_1	EXT_SOURCE_2
	EXT_SOURCE_3	FLAG_OWN_CAR	FLAG_OWN_REALTY	CNT_CHILDREN	CNT_FAM_MEMBERS

Selection Criteria

- External Credit Scores
  EXT_SOURCE_1,2,3
- The Client's Profile (Demographics & Stability):
  DAYS_BIRTH, NAME_EDUCATION_TYPE, NAME_FAMILY_STATUS, CNT_CHILDREN, CNT_FAM_MEMBERS
- The Financials
  NAME_INCOME_TYPE, AMT_INCOME_TOTAL, AMT_CREDIT, AMT_ANNUITY, AMT_GOODS_PRICE
- Assets
  NAME_HOUSING_TYPE, REGION_RATING_CLIENT, FLAG_OWN_CAR, FLAG_OWN_REALTY, DEF_30_CNT_SOCIAL_CIRCLE
  - + relatively complete compared with many property-detail columns that have very high missingness

Classifying Variables

Categorical

- NAME_FAMILY_STATUS
- NAME_HOUSING_TYPE
- NAME_INCOME_TYPE
- NAME_EDUCATION_TYPE
- OCCUPATION_TYPE
- REGION_RATING_CLIENT
- FLAG_OWN_CAR
- FLAG_OWN_REALTY

Numeric

- EXT_SOURCE_1
- EXT_SOURCE_2
- EXT_SOURCE_3
- DEF_30_CNT_SOCIAL_CIRCLE
- CNT_CHILDREN
- CNT_FAM_MEMBERS
- AMT_ANNUITY
- AMT_GOODS_PRICE
- AMT_INCOME_TOTAL
- AMT_CREDIT
- DAYS_BIRTH
- DAYS_EMPLOYED

Cleaning up Data

1.Missing Data

```
Missing data in our Driver Columns (%):
EXT_SOURCE_1                 56.381073
OCCUPATION_TYPE              31.345545
EXT_SOURCE_3                 19.825307
DEF_30_CNT_SOCIAL_CIRCLE      0.332021
EXT_SOURCE_2                  0.214626
AMT_GOODS_PRICE               0.090403
AMT_ANNUITY                   0.003902
CNT_FAM_MEMBERS               0.000650
dtype: float64
```

#Checking if missingess leads to default

```
EXT1_missing
False    0.074955
True     0.085195
Name: TARGET, dtype: float64
EXT3_missing
False    0.077665
True     0.093119
Name: TARGET, dtype: float64
OCC_missing
False    0.087851
True     0.065131
Name: TARGET, dtype: float64
```

→

- **Missing external score value applicants (EXT_SOURCE_1/EXT_SOURCE_3)**
  - higher default rates → may indicate weak external credit info → don't drop
- **Missing OCCUPATION_TYPE :**
  - lower default rate → could be pensioners → don't drop

2.Fix day-related anomalies

- **Replace days employed that are 365243 (1000 years) to NA**
- **Convert days employed/birth to years**

DAYS_EMPLOYED → YEARS_EMPLOYED

DAYS_BIRTH → AGE_YEARS

3. Fill missing numeric values with median

Categorical Variable Analysis

METHODOLOGY:

- Choice of plot: Barplot (nominal), Pointplot (ordinal)
- Reasoning:
  - “Target” is either only 0,1 → intuitive to get mean to get default rate, box/violinplot would be flat
  - too large of a dataset (307511 rows) for scatterplots

Nominal variables → Barplots

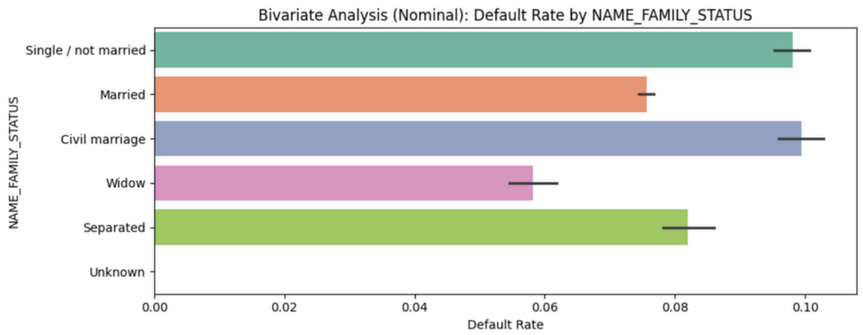
- NAME_FAMILY_STATUS
- NAME_HOUSING_TYPE
- NAME_INCOME_TYPE
- FLAG_OWN_CAR
- OCCUPATION_TYPE
- FLAG_OWN_REALTY

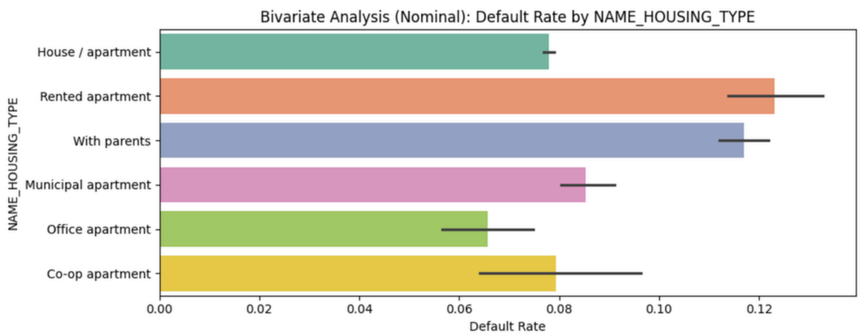
Ordinal variables → Pointplots

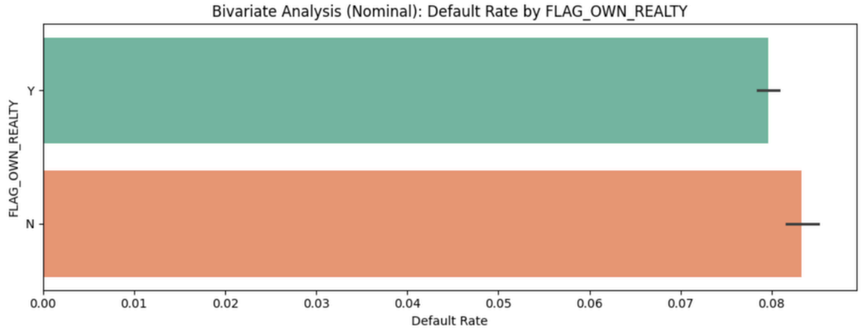
- NAME_EDUCATION_TYPE
- REGION_RATING_CLIENT

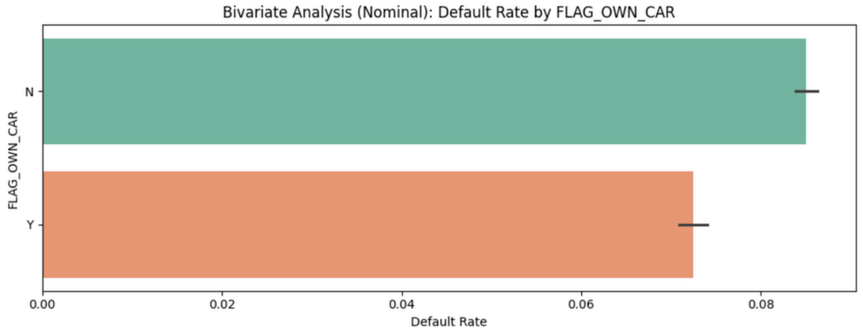
Categorical Variable Analysis

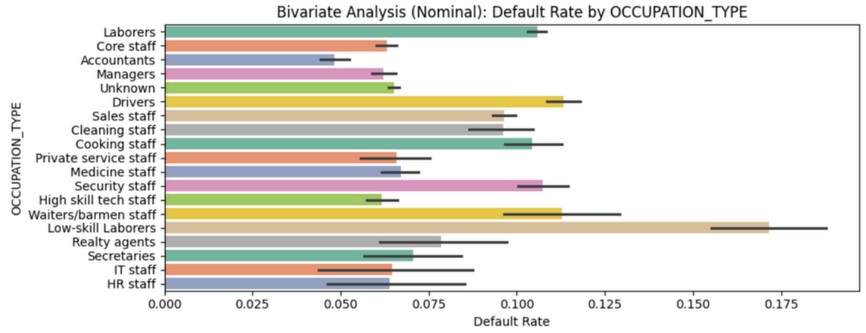
Nominal variables → Barplots

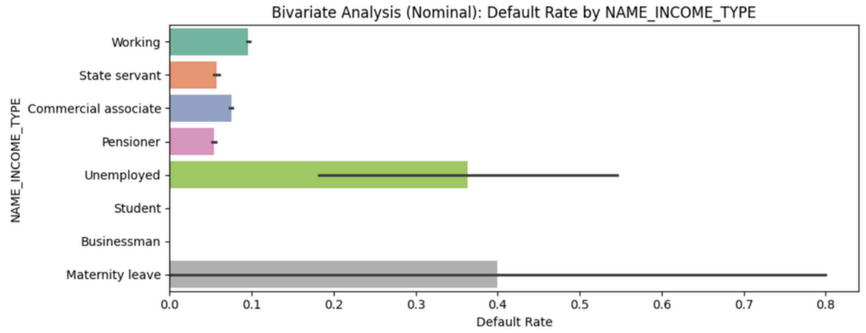












Categorical Variable Analysis

Key Takeaways for Nominal variables

High Risk Profiles

1. **Housing:** renting/ living with parents (~12% default rate)
2. **Family Status:** Single/Civil Marriage (~10% default rate)
3. **Occupation:** Low-skill laborers (spiking up to nearly 18%), Drivers/ Waiters/ Cooking/ Security staff also carry higher-than-average risk
4. **Income Type:** Unemployed/Maternity (~35-40% default rate)

Low Risk Profiles

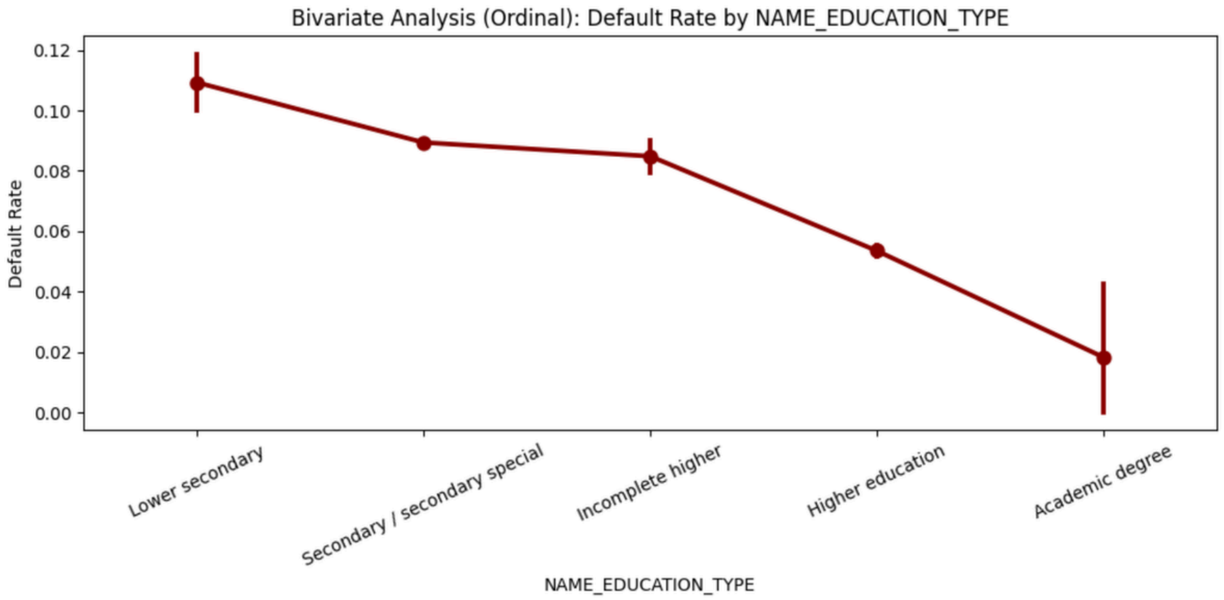
1. **Family Status: Widows actually have the lowest default rate.**
2. **Occupation:** Accountants, Private Service/Medicine/ High Skill Tech/IT/HR Staff are safer (5-6% default rate).
3. **Income Type:** Pensioners, State Servants

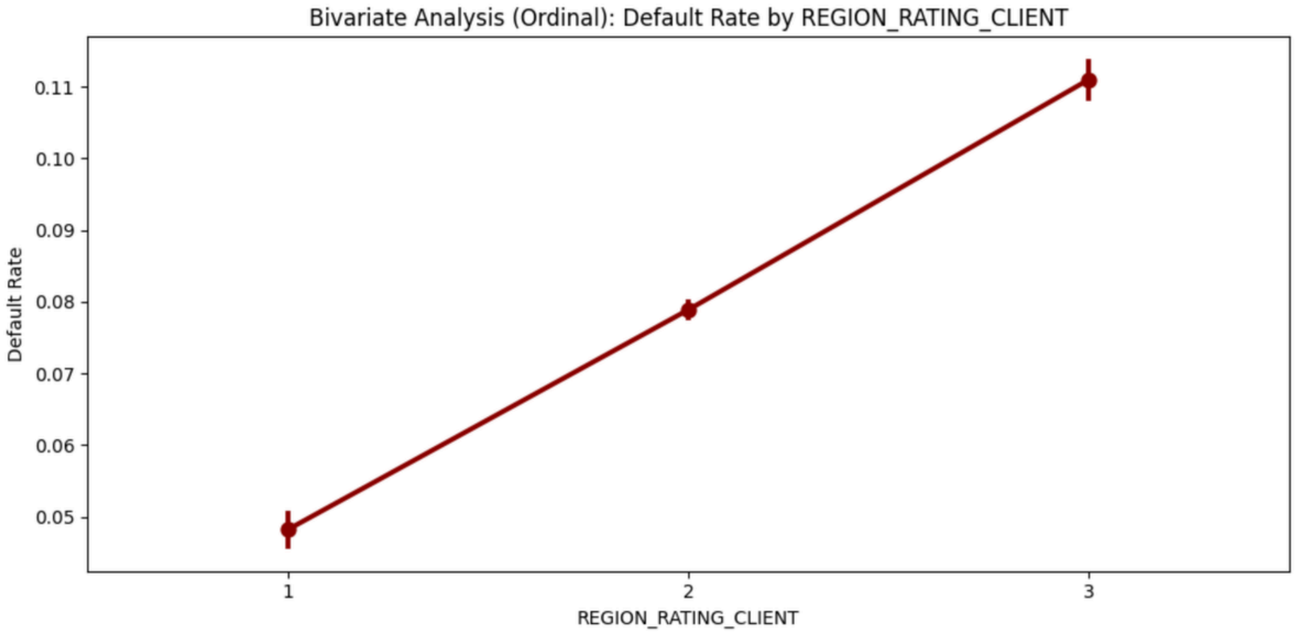
Cars vs. Real Estate

1. Realty: Whether one owns realty barely makes a difference in default rate
2. Car: However, owning a car has a noticeable drop in default rate

Categorical Variable Analysis

Ordinal variables → Pointplots





Categorical Variable Analysis

Key Takeaways for Ordinal variables

Inverse Relationship between education and default risk

1. **The High-Risk Group:** Applicants who only completed "Lower secondary" education are the riskiest (11% default rate)
2. **The Low- Risk Group:** The line drops onsistently with every degree earned. By the time an applicant reaches "Academic degree," their default risk drops to roughly 2%.
3. **Business Insight**: Investing in highly educated applicants is one of the safest bets the bank can make.

Inverse Relationship between Region Rating and default risk

1. Applicants from region 1 are reliable (~5% default rate), whereas region 3 are risky (~11% default rate)
2. **Business Insight**: Checking regional rating is an important factor.

Heatmap Analysis

Trying to understand how quantitative variables impact Target

METHODOLOGY:

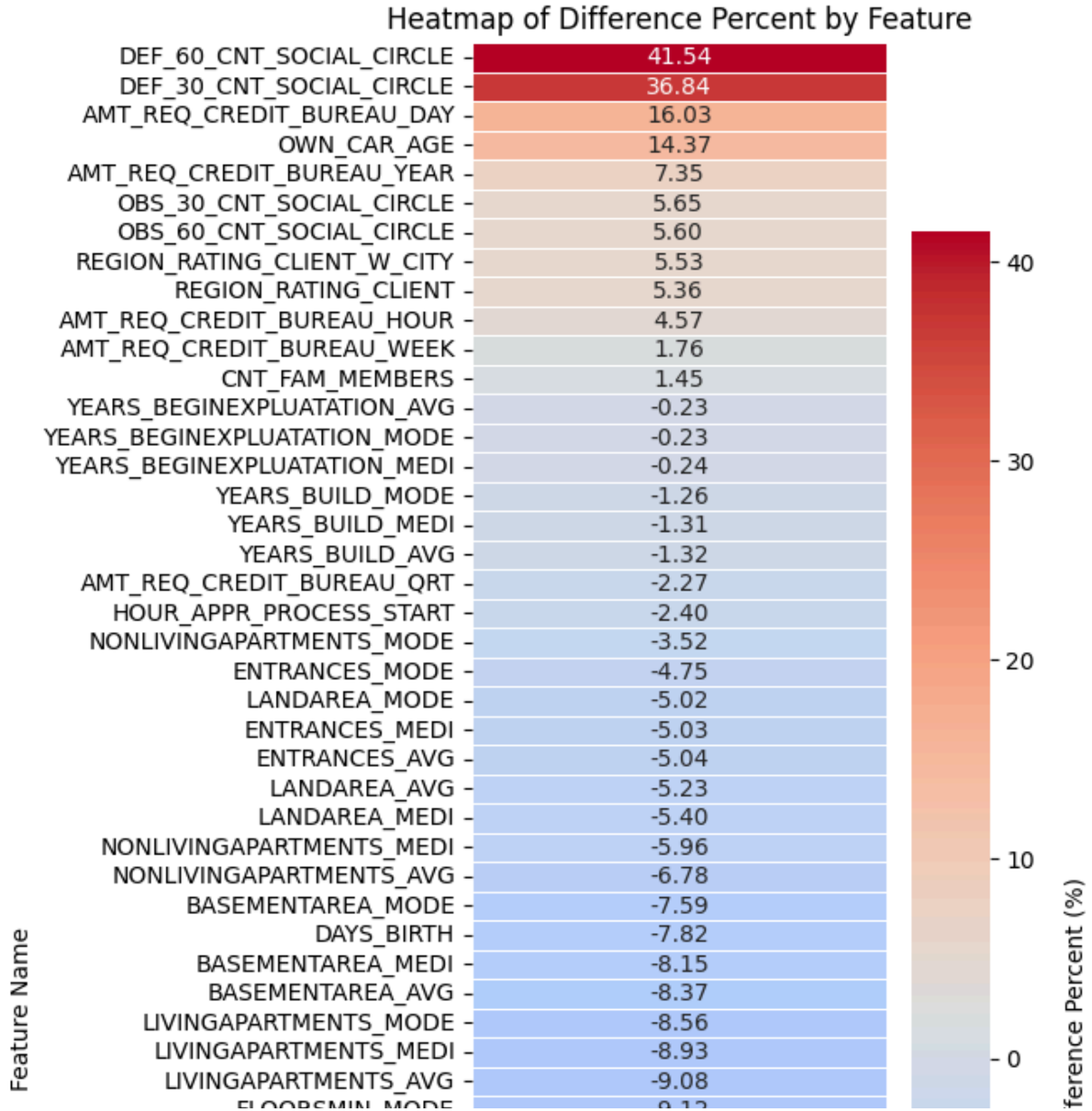
- Separate the data into two groups: TARGET=0 and TARGET=1
- Calculate the mean values for several different quantitative columns for both groups
- Calculate the difference of the means and divide it by the mean for both groups
- Look at the percentage difference for all the variables

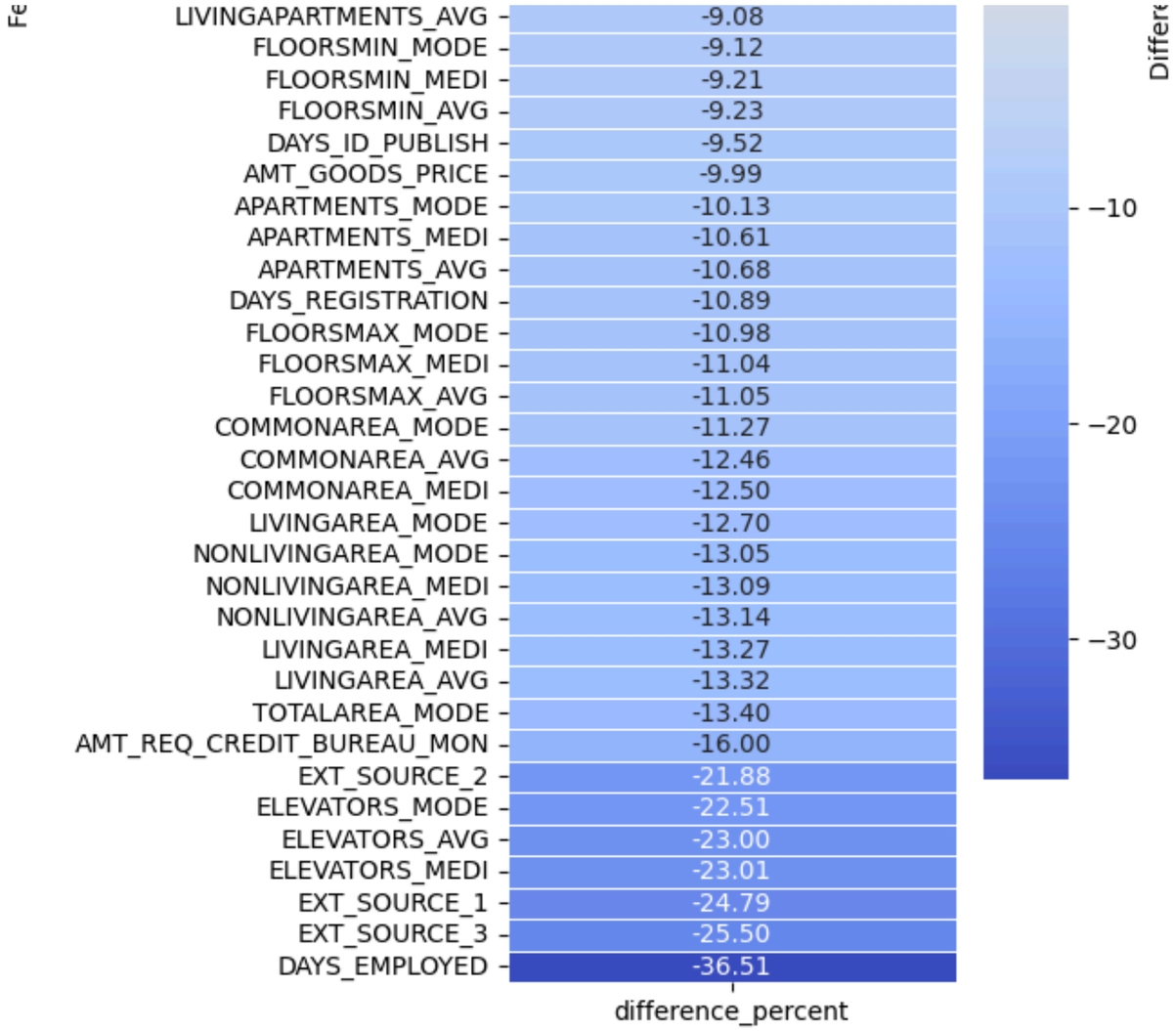
Heatmap Analysis

```
TARGET
0    542736.795003
1    488972.412554
Name: AMT_GOODS_PRICE, dtype: float64
The DIFFERENCE IS -9.99%

TARGET
0   -16138.176397
1   -14884.828077
Name: DAYS_BIRTH, dtype: float64
The DIFFERENCE IS -7.82%

TARGET
0    65696.146123
1    42394.675448
Name: DAYS_EMPLOYED, dtype: float64
The DIFFERENCE IS -36.51%

TARGET
0   -5029.941065
1   -4487.127009
Name: DAYS_REGISTRATION, dtype: float64
The DIFFERENCE IS -10.89%

TARGET
0   -3017.219788
1   -2732.099617
Name: DAYS_ID_PUBLISH, dtype: float64
The DIFFERENCE IS -9.52%
```

```
TARGET
0    1.415742
1    1.496147
Name: OBS_30_CNT_SOCIAL_CIRCLE, dtype: float64
The DIFFERENCE IS 5.65%

TARGET
0    0.139148
1    0.191980
Name: DEF_30_CNT_SOCIAL_CIRCLE, dtype: float64
The DIFFERENCE IS 36.84%

TARGET
0    1.398923
1    1.477672
Name: OBS_60_CNT_SOCIAL_CIRCLE, dtype: float64
The DIFFERENCE IS 5.60%

TARGET
0    0.096688
1    0.138247
Name: DEF_60_CNT_SOCIAL_CIRCLE, dtype: float64
The DIFFERENCE IS 41.54%

TARGET
0    0.006380
1    0.006672
Name: AMT_REQ_CREDIT_BUREAU_HOUR, dtype: float64
The DIFFERENCE IS 4.57%

TARGET
0    0.006914
1    0.008036
Name: AMT_REQ_CREDIT_BUREAU_DAY, dtype: float64
The DIFFERENCE IS 16.03%

TARGET
0    0.034315
1    0.034919
Name: AMT_REQ_CREDIT_BUREAU_WEEK, dtype: float64
The DIFFERENCE IS 1.76%
```

Heatmap Analysis





Heatmap Analysis

A high positive difference means that a TARGET value of 1 has a higher average of that variable. This means that the company should be putting a higher risk to a high number for that variable.

Strong Positive Risk Signals (Defaulters have much higher values)

DEF_60_CNT_SOCIAL_CIRCLE (+41.54%)
DEF_30_CNT_SOCIAL_CIRCLE (+36.84%)
AMT_REQ_CREDIT_BUREAU_DAY (+16%)
OWN_CAR_AGE (+14%)
REGION_RATING_CLIENT (+5%)
OBS_30_CNT_SOCIAL_CIRCLE (+5%)

Interpretations:

- Social circles are an important default interpreter.
- Applicants with other payment difficulties are more likely to default.
- Applicants who default have older cars on average.

Heatmap Analysis

A high positive difference means that a TARGET value of 1 has a higher average of that variable. This means that the company should be putting a higher risk to a high number for that variable.

Strong Negative Risk Signals (Defaulters have much lower values)

DAYS_EMPLOYED (–36.51%)

EXT_SOURCE_3 (–25.5%)

EXT_SOURCE_1 (–24.79%)

EXT_SOURCE_2 (–21.88%)

AMT_REQ_CREDIT_BUREAU_MON (–16%)

LIVINGAREA variables (–10 to –13%)

Interpretations:

- Defaulter had been working for a lower amount of time.
- Defaulters have smaller houses.
-

Historical Data Analysis

Credit Ratio Analysis

Methodology

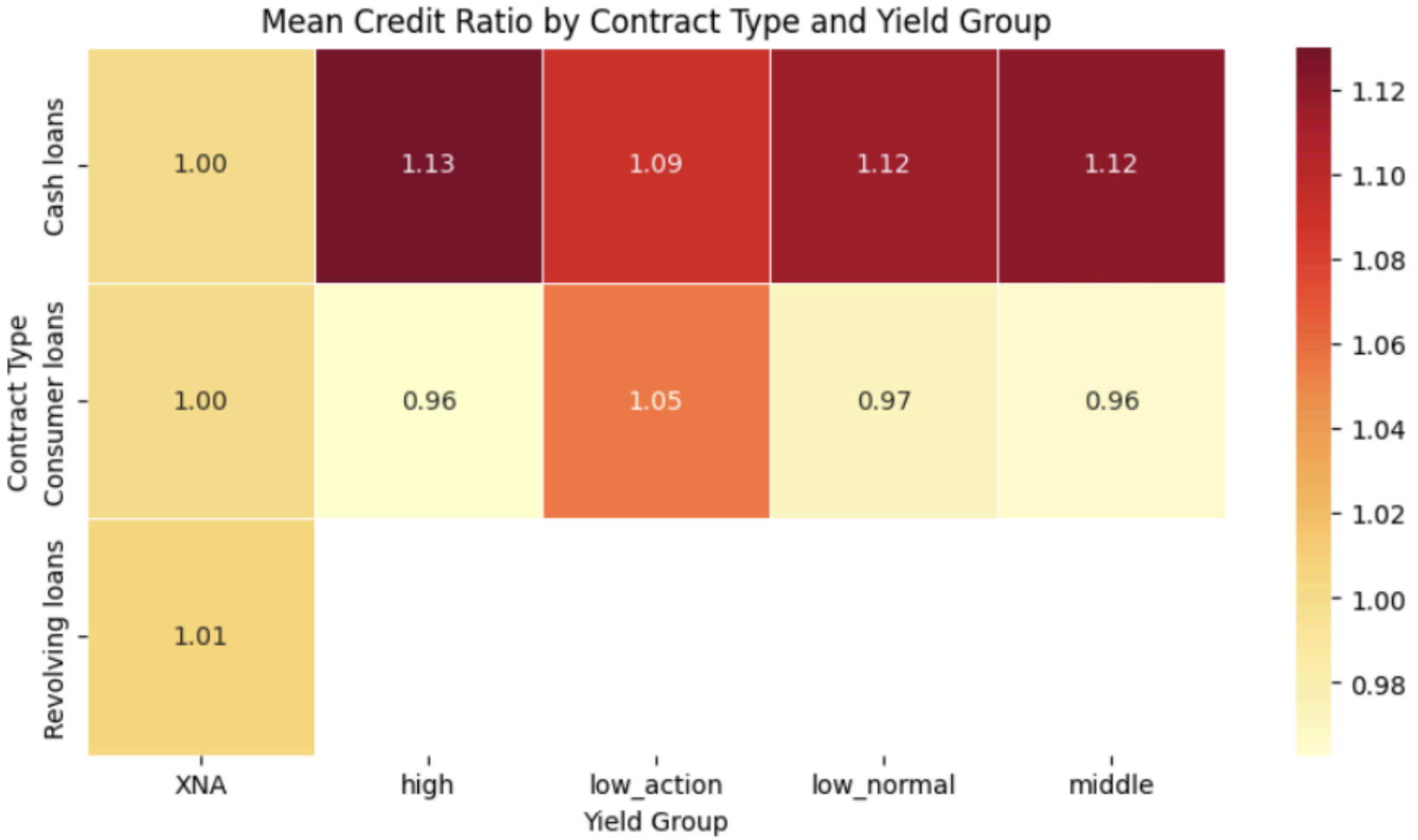
- Calculated a new column, CREDIT_RATIO by dividing AMT_CREDIT by AMT_APPLICATION to measure how much credit was granted relative to what was applied for
- Filtered out unreliable data by keeping only rows where CREDIT_RATIO was less than 2, removing division by zero errors and outliers
- Grouped the data by NAME_CONTRACT_TYPE and NAME_YIELD_GROUP and calculated the mean credit ratio for each combination
- Unstacked the result into a pivot table with contract types as rows and yield groups as columns
- Visualised the pivot table as a diverging heatmap centered at 1.0, where red indicates the bank granted more than applied for and yellow indicates less

New Variable

CREDIT_RATIO → LOGIC → AMT_CREDIT / AMT_APPLICATION

Historical Data Analysis

Credit Ratio Analysis



Historical Data Analysis

Credit Ratio Analysis Key Takeaways

Cash loans consistently receive more credit than requested

1. **Risk Signal**: Cash loan applicants receive 9–13% more credit than they apply for across all yield groups, suggesting the bank is actively upselling credit
2. **Business Insight**: Approving more credit than requested increases the repayment burden on the applicant, which could be a contributing factor to default risk

Consumer loans are more likely to receive less than applied for

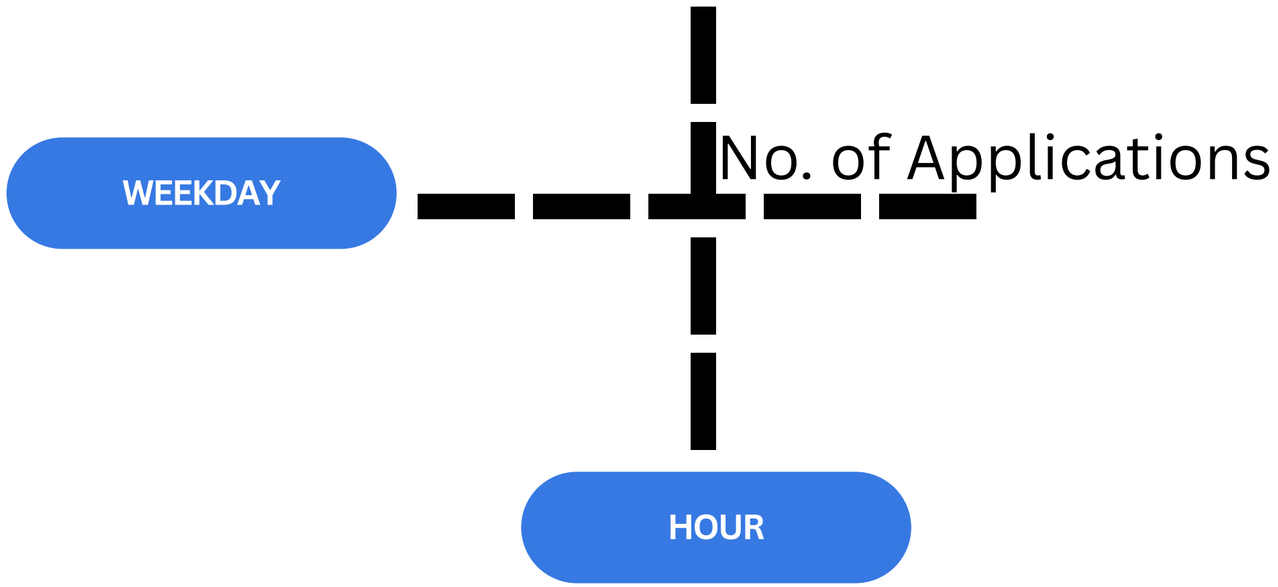
1. **Risk Signal**: Consumer loan ratios fall below 1.0 for most yield groups, meaning the bank is already being cautious, likely because this segment has a higher observed default rate
2. **Business Insight**: The bank's conservative approach to consumer loans may be justified, and yield group should be used as an active filter when evaluating consumer loan applications

Historical Data Analysis

Temporal Analysis of Application Volume

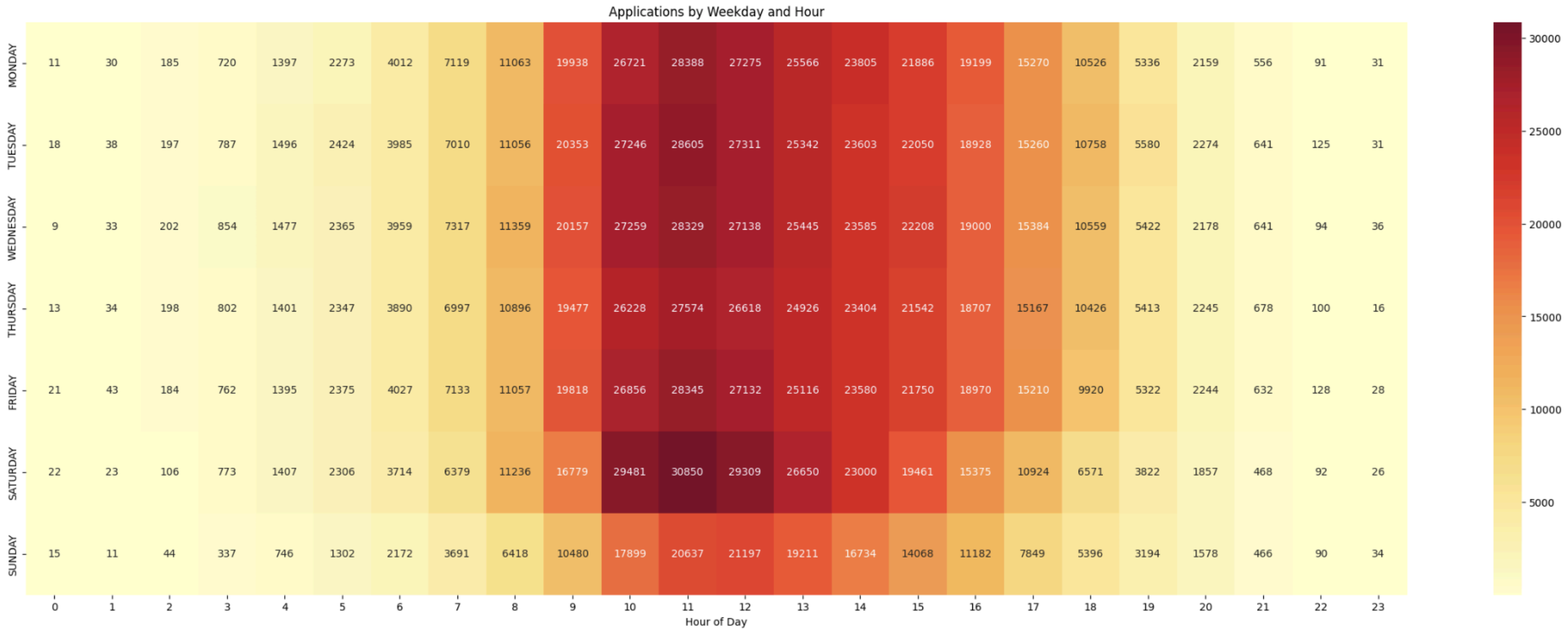
Methodology

- Grouped the data by WEEKDAY_APPR_PROCESS_START and HOUR_APPR_PROCESS_START and counted the number of applications for each combination
- Unstacked the result into a pivot table and reindexed rows to follow Monday–Sunday order
- Visualised as a heatmap where darker red indicates higher application volume



Historical Data Analysis

Temporal Analysis of Application Volume



Historical Data Analysis

Temporal Analysis of Application Volume Key Takeaways

Peak hours are consistent across all weekdays

1. **The Busiest Window**: Applications surge between 10am–12pm every day, with Saturday hitting the single highest count of 30,850 at 11am
2. **Business Insight**: Banks should ensure maximum staffing and system capacity during the 10am–12pm window across all days

Peak application hours are concentrated in the late morning

1. **The Risk Signal**: Saturday at 11am sees the highest volume (30,850 applications), and weekends overall show strong activity — these may be periods of less thorough review
2. **Business Insight**: High-volume periods with potentially stretched review capacity could lead to riskier loans being approved — flagging applications submitted during peak hours for additional scrutiny could reduce default rates

Thank You

- Minseo
- Onat
- Mauurya